

Drasil Onboarding New Project Members

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This document describes the onboarding process for new members of the Drasil team. The new members could be summer students, Masters students, PhD students, volunteers, etc. If a contributor is keen to begin, some of these tasks can be started before their official start date, although there is no expectation to do so.

Below you will find the following: a summary of your work colleagues (Section 1), practical information for summer research assistants (Section 2), a summary of the relevant repositories (Section 3), a list of initial tasks (Section 4), advice from previous contributors (Section 5) and a list of “to do”s for Drs. Carette and Smith (Section 6).

The details of your initial tasks are in Section 4. The first tasks listed are related to GitHub and GitLab access. As described in Section 4, please do the following as your first tasks:

- send Drs. Carette and Smith your GitHub username
- register for GitLab (if you haven’t already done so)

The most important getting-started advice is to remind you of the importance of communication. We’ll do our best to communicate our requirements and expectations. You should likewise do your best to communicate when you are confused, frustrated, bored etc. Our goal is to keep you busy with a fun and rewarding experience.

1 Introductions

In addition to [Dr. Jacques Carette](#) and [Dr. Spencer Smith](#) (smiths@mcmaster.ca), here is a list of the current (as of the date of generating this document) members of the Drasil team:

- Jason Balaci, PhD candidate
- Samuel Crawford, MASc candidate
- Jack Wyand, Summer research assistant
- Sarra Soussia, Mitacs research assistant

2 Summer Research Assistant Practical Information

Start date: Monday, May 5, 2023

End date: Variable

Workspace: ITB/236 (to be verified). You will need a proximity card to access your office. You will need to go to JHE 216A (Engineering Support Services ([The Hub](#))) to get your card. A deposit is required for the card. If there are any problems, please coordinate with the Departmental Administrator (Mr. Matt Vonk).

First day: Come to ITB/167 to see Dr. Smith at 9:30 am. We won't have a meeting at this time, just a brief exchange about the practical details of your first day. You will want to get your access card, claim a spot in your office, and begin reviewing these onBoarding instructions.

Working from home: Once we get our summer work rhythm established, you can potentially work from home on some days, but please discuss this with Drs. Carette and Smith.

Hours: 35 hours per week, 7 hours per day (Monday to Friday) with a one-hour unpaid lunch. You can take a half-hour lunch if you prefer, but a lunch break is required. We will maintain a regular work day. There is some flexibility on the start time. Any time between 8:30 am and 9:30 am is fine. In some cases, you may need to alter your work schedule for personal reasons. This is fine, but we need to discuss the proposed alternatives.

Please use a spreadsheet to keep track of your hours and the tasks that you spend your time on. For the first week, please e-mail the spreadsheet to Drs. Smith and Carette at the end of each day. The purpose of communicating this information is to help advise and understand; it is not to "check up on you."

Overview meeting: TBD. Dr. Smith, and Jason Balaci will [present](#) an overview of Drasil's purpose and its implementation. Dr. Carette will attend if available.

“All hands” meetings: TBD [First meeting](#) of the full summer research Drasil team.

For meetings, we prefer in-person whenever feasible.

3 Repos

We use several repos for our work. They are listed below in roughly in order of importance. In some cases, you will need to create an account or access will have to be given. The specific access-related tasks are given in the next section (Section [4](#)).

3.1 GitHub

3.1.1 Drasil

- [Drasil Repo](#)
- public repo
- the source code and documentation for Drasil
- any code or documentation you write on Drasil will be here
- you cannot push to “main”
- all contributions will be done through pull requests
- this is the repo where you will be doing most of your work
- you will need to be added as a contributor (see Section [4](#))

3.1.2 Drasil Generated Case Studies and Documentation

- [Drasil Case Studies and Documentation](#)
- not actually a repo, but generated from Drasil
- the automated versions of the Drasil case studies are built frequently and pushed to this web-page

- the Haskell dependency graphs are also provided here
- the [documentation](#) for Drasil is available
- [package dependency graphs](#) are available at the bottom of this web page

3.2 GitLab (CAS server)

3.2.1 Publications

- [Publications](#)
- private (within CAS) repo
- bibliographic information (in BibTeX) for papers and other resources relevant to our project
- pdf versions of papers that are hard to find online
- when you create a bib file, look here (in the `References.bib` file) first to see if the bib data is already available
- if you find a new reference, please add it to the `References.bib` file, along with a pdf version, if you don't have a link to an online version
- citations should be named using the Author Year style. For one or two authors their last names are listed and then the year. For more than two authors, the [first author's last name is listed followed by Et al.](#)
- contributors can push to "main"
- you need to be added as a contributor to this repo

3.2.2 Software Engineering Course Notes

- [SE2AA4/CS2ME3 Course Notes](#)
- public repo
- on some occasions, we may refer to some of the concepts or technology from software engineering; this repo might be referenced in those situations

- you cannot push, but you can do a pull request, if necessary

3.2.3 Software Engineering for Science

- [se4sc repo](#)
- private repo
- resources
- grad student and undergrad student work
- paper drafts
- research proposal drafts
- any documents you write that aren't part of Drasil will be put in this repo
- contributors can push to “main”
- depending on your work you might not need this resource; we will let you know when it is relevant
- in case it is relevant, you will be added as a contributor to this repo

4 Initial Tasks

If you have questions or challenges while completing the steps below, please make a record of your challenge. We are always working to improve our onboarding instructions and contributor's guide. Please let us know of any problems with the documentation so that can address the problem in the future. There is a good chance we'll ask you to update the documentation, so the better your notes, the easier the task will be.

Once you get settled, you can begin with the tasks listed here. These tasks should be done in roughly the order listed. Some of the later tasks do not need to be done when you start (like learning LaTeX). They are tasks you can return to throughout the summer when you need something to do or when you are feeling like a change of pace, or waiting for feedback on your work.

1. Verify you can access all GitLab accounts on the CAS server. You can access GitLab at the [sign in page](#). For CAS students, you can follow the instructions on the screen to create an account, if you haven't already. For the nonCAS students, we'll work on getting you added by asking Derek (Sys Admin for CAS). Please let Dr. Smith know if you need us to request an account for you.
2. GitHub account. If you do not have one, please create one. Send your account username to Dr. Smith and Dr. Carette. Verify that you can access all of the GitHub repos listed above.
3. Consent to Provide Limited Personal Information about Highly Qualified Personnel (HQP) to NSERC. Dr. Smith will send a separate e-mail about this.
4. Connect with Drs. Carette and Smith over [Linked-In](#)
5. Familiarize yourself with Drasil, review the quick start guide and set up your new Drasil workspace. The relevant links are as follows:
 - [wiki](#)
 - [wiki - what is Drasil?](#)
 - [quick start](#)
 - [new workspace setup](#)

The last link is particularly practical and useful. You'll want to follow the new workspace setup instructions to have a sane build environment. If you have any problems setting up your Drasil workspace, post an issue on GitHub. The issue should include the details of your OS, what you have tried, and any relevant screenshots.

6. Learn the basics of git (if you don't already know them). An overview of git can be found from the following resources:
 - [Capstone tutorial by Sam Crawford on Teams](#)
 - [Capstone Tutorial Info](#)
 - [Capstone Cheat Sheet \(also by Sam\)](#)
 - [2AA4/2ME3 Tutorial](#)

- [Git2Know for Drasil](#)
7. Review the [Contributor's Guide](#)
 Issue tracking is also described in the Contributor's Guide. Please follow the issue tracking guidelines. You should also review the existing issues in Drasil, especially those that are currently open.
 8. Once you have reviewed the material on git and contributing, please complete the [Contributor's Test](#). Complete the version without answers and then compare your answers to the correct ones. If you have any questions, please ask.
 9. We will be using Haskell this summer. A good introductory text for Haskell is available at [Learn You a Haskell for Great Good](#). You should have installed Haskell as part of the [New Workspace Setup](#). Other Haskell resources include:
 - Coursera course on [Programming Languages, Part A](#) (The course isn't specifically on Haskell, but the languages used are similar enough)
 - McMaster Univ [Comp Sci 1JC3 online lectures](#)
 - McMaster Univ [Comp Sci 1JC3 online tutorials](#)
 10. Start working on any issues assigned to you. We usually start with the issues labelled "newcomers". Please do not hesitate to ask for help with (a) solving issues (b) finding new issues. You are also welcome to explore the issue tracker to find other issue that interest you.
 11. Read and review the following Drasil-related papers. These papers describe the Drasil framework and the companion Generic Object Oriented Language (GOOL).
 - [What If \(we generate everything\)?](#) (also known as the story of Drasil)
 - [Drasil Position Paper](#)
 - [Generating Software for Well-Understood Domains](#)
 - [GOOL paper](#), if relevant to your work, you can look at the.

12. If you are new to software development tools, you will want to review the lessons on [Software Carpentry](#). In particular, the lessons on the Unix Shell and make would be a good start.
13. Learn basics of LaTeX. We will be using LaTeX for our written documents. You can find an overview in the [SE2AA4/CS2MED tutorial](#).

5 Advice from Previous Year's Research Assistants

- “Don’t be afraid to ask questions.”
- “You rarely have to implement at a low level, chances are many of the things you want to implement have already been implemented (i.e. check existing modules and packages there might be similar functions).”
- [Contributor's Guide notes for summer research students](#)

6 Drs. Carette and Smith To Do for New Contributors

- add contributor to Drasil repo (GitHub)
- add contributor to se4sc and pub repos (GitLab)
- request CAS accounts for non-CAS contributors
- add contributor to Discord
- schedule meetings
- assign initial Drasil issues
- request Consent to Provide Limited Personal Information about Highly Qualified Personnel (HQP) to NSERC e-mail
- add contributors to Teams (Rsch Stdnt Meeting Team)

7 Offboarding

When your “official” time with the Drasil project is wrapping up, please consider the items in the following checklist:

- For all your open branches, if they are unsalvageable or not useful, please delete them. Otherwise, please file a draft PR, explaining the state and nature of the work. If possible, explaining how others may continue it.
- For all your open issues and/or pull requests, please discuss the state of the work. It would be appreciated if you could leave it in a state where another summer student could read your comments and pick up where you left off.
- Ensure your names is added to the [CITATION.cff](#) file.

When your time as a grad student or employee is over, we encourage you to stay as engaged in the Drasil project as your time and interest allows.